

Mode 9 — Square (centered / opt)

DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → DP8 (4×4) → centered add-and-square

Step 1 — DP8 (16×16)

$$\text{DP8 } (16 \times 16) = \sum_{i=0}^7 a_i b_i$$

Step 2 — DP8 (16×8)

$$\text{DP8 } (16 \times 16) = 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 } (16 \times 8)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 } (16 \times 8)}$$

Step 3 — DP8 (16×4)

$$\text{DP8 } (16 \times 16) = 2^{12} \underbrace{\sum_{i=0}^7 a_i b_i^{hh}}_{\text{DP8 } (16 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hl}}_{\text{DP8 } (16 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{lh}}_{\text{DP8 } (16 \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{ll}}_{\text{DP8 } (16 \times 4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned} \text{DP8 } (16 \times 16) = & 2^{20} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hh}}_{\text{DP8 } (8 \times 4)} + 2^{16} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hl}}_{\text{DP8 } (8 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hh}}_{\text{DP8 } (8 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lh}}_{\text{DP8 } (8 \times 4)} \\ & + 2^8 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hl}}_{\text{DP8 } (8 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{ll}}_{\text{DP8 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lh}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{ll}}_{\text{DP8 } (8 \times 4)} \end{aligned}$$

Step 5 — DP8 (4×4)

$$\begin{aligned} \text{DP8 } (16 \times 16) = & 2^{24} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{20} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{20} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{16} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{hl}}_{\text{DP8 } (4 \times 4)} \\ & + 2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hh}}_{\text{DP8 } (4 \times 4)} + 2^{16} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{lh}}_{\text{DP8 } (4 \times 4)} \\ & + 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{hl}}_{\text{DP8 } (4 \times 4)} + 2^{12} \underbrace{\sum_{i=0}^7 a_{H,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=0}^7 a_{L,i}^{hi} b_i^{ll}}_{\text{DP8 } (4 \times 4)} \\ & + 2^8 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{lh}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i}^{lo} b_i^{ll}}_{\text{DP8 } (4 \times 4)} \end{aligned}$$

Step 6 — Centered add-and-square (subtract 8 from unsigned nibbles; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles)

$$\text{DP8 } (16 \times 16) = 2^{24} \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{hi})^2 - \sum_{i=0}^7 (b_i^{hh})^2 \right)}_{\text{DP8 } (4 \times 4)}$$

$$\begin{aligned}
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{20} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{hl} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 - \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + b_i^{hh})^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 8)^2 - \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + b_i^{hh})^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 8)^2 - \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{16} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{lh} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 - \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{hl} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^{12} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 - \sum_{i=0}^7 (b_i^{ll} - 8)^2 + 512 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 - \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{lh} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

$$\begin{aligned}
& + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{ll} - 8))^2 - \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 - \sum_{i=0}^7 (b_i^{ll} - 16)^2 + 2048 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 7 — All components

$$\begin{aligned}
\text{PE} &= 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi} + b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{hl} - 8))^2 + 2^{16} \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{hl} - 8))^2 \\
&+ 2^{16} \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + b_i^{hh})^2 + 2^{12} \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + b_i^{hh})^2 + 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{lh} - 8))^2 + 2^{12} \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{lh} - 8))^2 \\
&+ 2^{12} \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{hl} - 8))^2 + 2^8 \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{hl} - 8))^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} + (b_i^{ll} - 8))^2 + 2^8 \sum_{i=0}^7 ((a_{L,i}^{hi} - 8) + (b_i^{ll} - 8))^2 \\
&+ 2^8 \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{lh} - 8))^2 + 2^4 \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{lh} - 8))^2 + 2^4 \sum_{i=0}^7 ((a_{H,i}^{lo} - 8) + (b_i^{ll} - 8))^2 + \sum_{i=0}^7 ((a_{L,i}^{lo} - 8) + (b_i^{ll} - 8))^2 \\
\alpha &= 2^{24} \sum_{i=0}^7 (a_{H,i}^{hi})^2 + 2^{20} \sum_{i=0}^7 (a_{L,i}^{hi} - 8)^2 + 2^{20} \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 + 2^{16} \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 \\
&+ 2^{16} \sum_{i=0}^7 (a_{H,i}^{lo} - 8)^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{lo} - 8)^2 + 2^{16} \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 + 2^{12} \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 + 2^{12} \sum_{i=0}^7 (a_{H,i}^{hi} - 8)^2 + 2^8 \sum_{i=0}^7 (a_{L,i}^{hi} - 16)^2 \\
&+ 2^8 \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 + 2^4 \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 + 2^4 \sum_{i=0}^7 (a_{H,i}^{lo} - 16)^2 + \sum_{i=0}^7 (a_{L,i}^{lo} - 16)^2 \\
\beta &= 2^{24} \sum_{i=0}^7 (b_i^{hh})^2 + 2^{20} \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 2^{20} \sum_{i=0}^7 (b_i^{hl} - 8)^2 + 2^{16} \sum_{i=0}^7 (b_i^{hl} - 16)^2 \\
&+ 2^{16} \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 2^{12} \sum_{i=0}^7 (b_i^{hh} - 8)^2 + 2^{16} \sum_{i=0}^7 (b_i^{lh} - 8)^2 + 2^{12} \sum_{i=0}^7 (b_i^{lh} - 16)^2 \\
&+ 2^{12} \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2^8 \sum_{i=0}^7 (b_i^{hl} - 16)^2 + 2^{12} \sum_{i=0}^7 (b_i^{ll} - 8)^2 + 2^8 \sum_{i=0}^7 (b_i^{ll} - 16)^2 \\
&+ 2^8 \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2^4 \sum_{i=0}^7 (b_i^{lh} - 16)^2 + 2^4 \sum_{i=0}^7 (b_i^{ll} - 16)^2 + \sum_{i=0}^7 (b_i^{ll} - 16)^2 \\
C &= 0 + 536870912 + 536870912 + 134217728 + 33554432 + 2097152 + 33554432 + 8388608 \\
&+ 8388608 + 524288 + 2097152 + 524288 + 524288 + 32768 + 32768 + 2048 = 1297680384
\end{aligned}$$